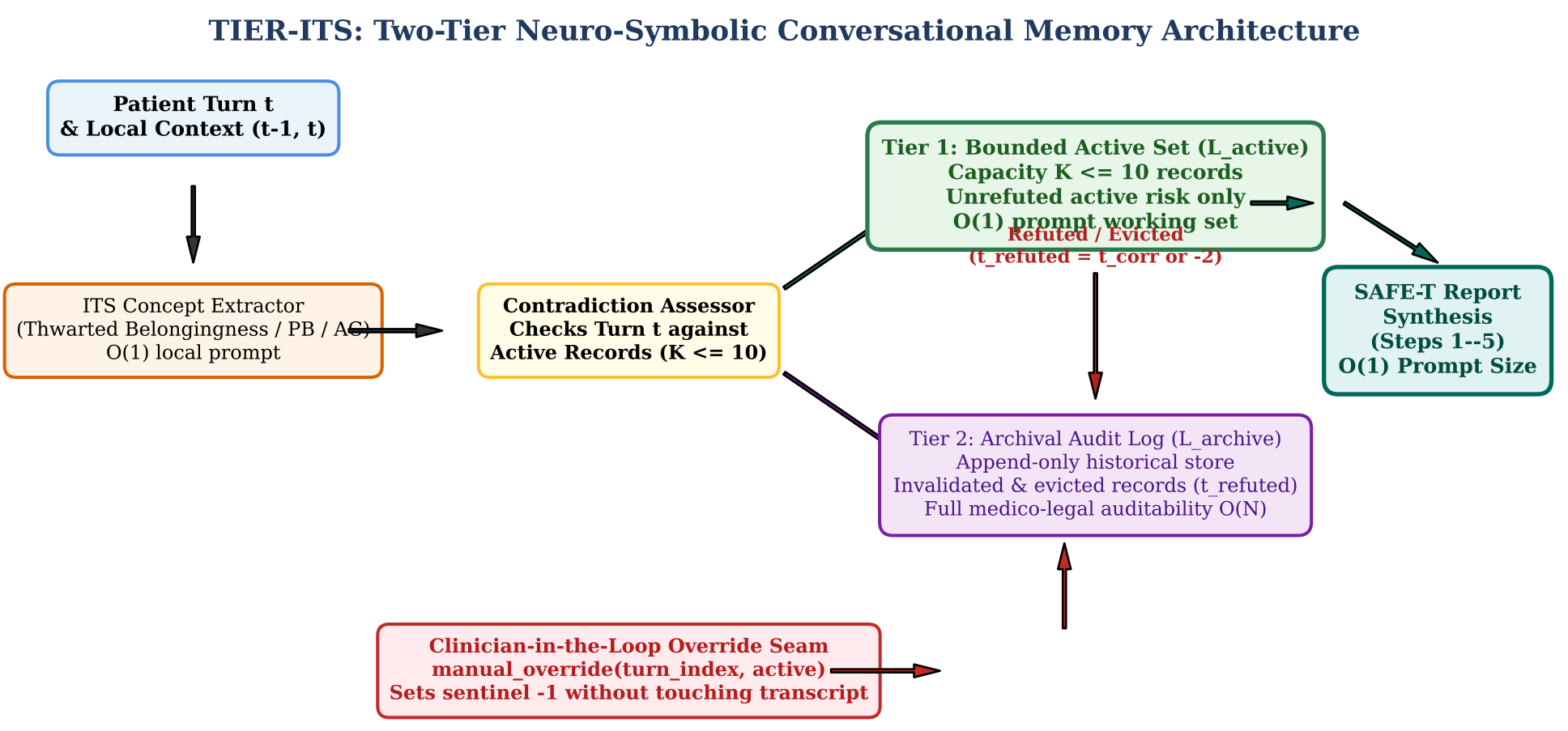
TIER-ITS: Two-Tier Neuro-Symbolic Conversational Memory Architecture
Patient Turn t
& Local Context (t-1, t)
ITS Concept Extractor
(Thwarted Belongingness / PB / AC)
O(1) local prompt
Contradiction Assessor
Checks Turn t against
Active Records (K <= 10)
Tier 1: Bounded Active Set (L_active)
Capacity K <= 10 records
Unrefuted active risk only
O(1) prompt working set
Refuted / Evicted
(t_refuted = t_corr or -2)
SAFE-T Report
Synthesis
(Steps 1--5)
O(1) Prompt Size
Tier 2: Archival Audit Log (L_archive)
Append-only historical store
Invalidated & evicted records (t_refuted)
Full medico-legal auditability O(N)
Clinician-in-the-Loop Override Seam
manual_override(turn_index, active)
Sets sentinel -1 without touching transcript